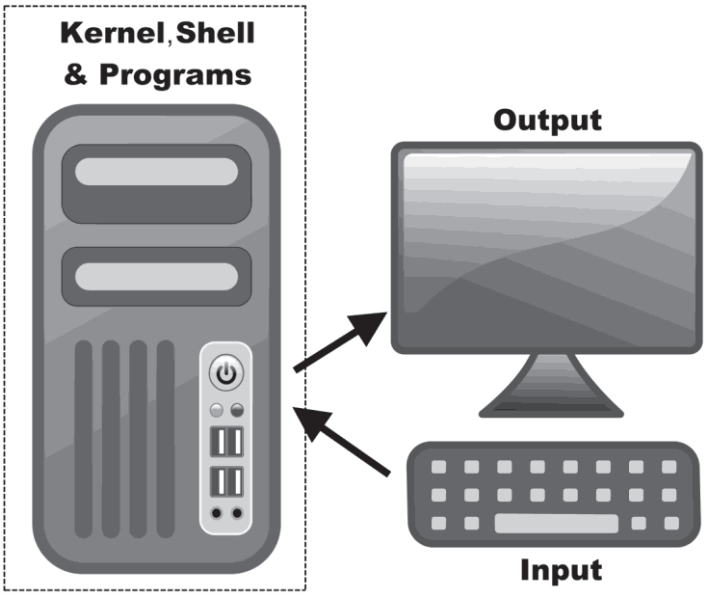
Section 1: Overview of Unix, Linux and BSD Architecture

The Kernel

At the heart of every Unix, Linux, and BSD system is the kernel. The kernel provides a layer between the computer hardware and user applications. When applications need to display output to a screen or write files to the disk, the kernel facilitates these actions. Think of the kernel as the conductor in a symphony; it directs all the actions of the operating system.



Overview of the Kernel

When you type a command on the keyboard, the computer hardware receives the signals and transfers them to the kernel. The kernel recognizes the key strokes and draws them on the display. Once the typed command is executed, the program will utilize the kernel to request processor, memory, and other resources in order to perform the requested task.

Popular Versions of Unix, Linux, and BSD

There are many different Unix, Linux, and BSD distributions. The tables below list the most popular systems in use today.

BSD Systems

OS	Description
Darwin	BSD-based core of the Mac OS X operating system
FreeBSD	User friendly BSD-based operating system
OpenBSD	Security focused BSD-based operating system
NetBSD	Highly portable BSD-based operating system

Linux Systems

OS	Description
Red Hat Enterprise Linux	Popular commercial Linux distribution created by Red Hat, Inc.
Fedora	Free community-based distribution that serves as a test bed for new technologies that feed into Red Hat Enterprise Linux
CentOS	Free binary-compatible clone of Red Hat Enterprise Linux
Debian	Popular non-commercial Linux distribution
Ubuntu	User-friendly Linux distribution based on Debian
SuSE Enterprise Linux	Commercial Linux distribution created by Novell, Inc.
openSUSE	Free community-based distribution based on SUSE Enterprise Linux

Unix Systems

OS	Description
AIX	IBM's proprietary Unix platform
HP-UX	Hewlett Packard's proprietary Unix platform
Solaris	Sun Microsystems' proprietary Unix platform
OpenSolaris	Open source version of the Solaris operating system

Note	*Other notable Unix platforms include SGI IRIX, Compaq Tru64, and SCO UnixWare/OpenServer; however, these systems are no longer widely used or supported.*

Unix, Linux, and BSD Directory Structure

Most Unix, Linux, and BSD systems share a similar directory structure. The table below lists the main directories commonly used across all platforms.

Directory	Purpose
`/`	Root (top level) directory
`/boot`	Linux kernel and boot loader files

`/stand`
`/unix`
`/vmunix`
`/kernel`
	Unix kernel directories

`/bin`
`/usr`
`/usr/bin`
	Core binary programs for normal users

`/sbin` `/usr/sbin`	Administrative programs primarily used by the root user
`/opt`	Optional add-on/third-party software
`/etc`	System configuration files
`/home`	User home directories
`/root`	Home directory of the root user
`/lib`	Shared libraries used by various programs
`/media` `/mnt`	Mounted local and remote file systems
`/var`	Variable data (such as system logs) that changes often
`/tmp`	Temporary files used by various programs
`/dev`	Device files
`/sys`	Virtual device driver files used by the kernel
`/proc`	Virtual parameter and informational files used by the kernel
`/lost+found`	Files that have been recovered (after a system failure)

Unix, Linux, and BSD directory structure

These directories are organized in a manner which separates various components of the operating system in an easy to manage file structure. The `/home` directory is where users spend most of their time. Files outside of this directory are usually secured so that only the root user can edit or delete them. This protects vital programs and configuration files and prevents users from accidently damaging the core operating system. It also helps secure the system by preventing malicious programs from being able to compromise critical files and settings.

Important Files

In addition to sharing a similar directory structure, most systems will also have the important files listed in following table.

File	Purpose
/etc/passwd	User account settings
/etc/shadow	Encrypted passwords for user accounts
/etc/group	Group settings
/etc/fstab	Local file system settings
/etc/mtab	Mounted file systems
/etc/inittab	System startup settings
/etc/exports	NFS share settings
/etc/hosts	Static name resolution file
/etc/hostname	System host name setting
/etc/hosts.allow	Network hosts allowed to connect to the system
/etc/resolve.conf	DNS settings
/etc/hosts.deny	Network hosts not allowed to connect to the system
/etc/issue	Message displayed at shell login
/etc/issue.net	Remote login message
/etc/motd	Message displayed after logging in
/etc/profile	Shell environment settings
/etc/shells	Valid shells permitted to login
/etc/sudoers	Users allowed to use the **su** command (discussed on page 90)
/var/log/messages	Kernel messages log file
/var/log/wtmp	Current user login log file
/var/log/lastlog	User login history log file
/var/log/kern.log	Kernel log file
/var/log/syslog	Syslog messages log file

Important files

This is just a brief list of the most commonly used files shared across most platforms. As a general rule, configuration files are located in the `/etc` directory and system logs are stored in `/var/log`. As you become more familiar with Unix, Linux, and BSD systems you will begin working with these files to manage system configuration and troubleshoot problems.

Common Unix, Linux, and BSD File Systems

Unix, Linux, and BSD share support for several types of file systems. The table below provides a matrix of the most common file systems and their supported platforms.

	AIX	HP-UX	Solaris	BSD	Linux
EXT2-4	Partial	No	Partial	Partial	Yes
JFS (IBM)	Yes	No	No	No	Yes
HFS+	No	No	No	No	Partial
UFS	No	Yes	Yes	Yes	Partial
VxFS	Yes	Yes	Yes	No	Yes
ZFS	No	No	Yes	Yes	Yes

File system support matrix

EXT versions 2 through 4 are file systems used by the Linux operating system. Currently, ext3 is the most widely used file system on Linux distributions. The recently released ext4 file system has many new features and performance benefits over ext3. It will become the default file system for most Linux systems within the next few years. Versions before ext3 are considered obsolete and are rarely.

JFS is IBM's journaling file system primarily used on the AIX operating system.

HFS+ is Apple's file system used on their Unix-based Mac OS X operating system. HFS+ isn't widely supported by other operating systems but it can be mounted as read-only on Linux systems.

UFS is a Unix file system used primarily by BSD distributions.

VxFS is a file system developed by Veritas Software Corporation (now known as Symantec) and used on HP-UX systems. It is also supported on several other Unix and Linux systems although it is rarely used outside of HP-UX.

ZFS is a new and highly robust file system developed by Sun Microsystems. It is primarily used on the Solaris operating system but also has limited support for some Linux and BSD systems.

Devices

Every Unix, Linux, and BSD system has a directory that is known as `/dev`. This directory contains device files (sometime referred to as special files) that represent hardware installed on the system such as `/dev/mouse` and `/dev/cdrom`.

```
# ls -l /dev/
crw-rw----+ 1 root audio  14,  12 2010-04-22 16:30 adsp
crw-------  1 root video  10, 175 2010-04-22 11:30 agpgart
crw-rw----+ 1 root audio  14,   4 2010-04-22 16:30 audio
crw-rw----  1 root root   10,  59 2010-04-22 11:30 binder
drwxr-xr-x  2 root root       700 2010-04-22 16:30 block
drwxr-xr-x  3 root root        60 2010-04-22 11:30 bus
lrwxrwxrwx  1 root root         3 2010-04-22 16:30 cdrom -> sr0
lrwxrwxrwx  1 root root         3 2010-04-22 16:30 cdrw -> sr0
drwxr-xr-x  2 root root      3220 2010-04-22 16:30 char
crw-------  1 root root    5,   1 2010-04-23 07:36 console
lrwxrwxrwx  1 root root        11 2010-04-22 16:30 core -> /proc/kcore
crw-rw----  1 root root   10,  58 2010-04-22 11:30 cpu_dma_latency
drwxr-xr-x  6 root root       120 2010-04-22 11:30 disk
drwxr-xr-x  2 root root        80 2010-04-22 11:30 dri
crw-rw----+ 1 root audio  14,   3 2010-04-22 16:30 dsp
lrwxrwxrwx  1 root root         3 2010-04-22 16:30 dvd -> sr0
...
```

Example listing of devices in the /dev directory

Within the `/dev` directory there can be several hundred files. Most of these files are of little use in everyday activities. As the system administrator of a Unix or Linux system you will primarily be working with the device files that represent your disk drive(s). Knowing the path to your hard drive is important when working with commands related to file systems. More on this topic is covered in Section 10 of this guide.

Note	*Device naming conventions used in* `/dev` *vary across each distribution.*

Within `/dev`, there are also a few helpful pseudo-devices. Pseudo-devices do not represent actual hardware and exist only to perform a specific task. The table below describes the most commonly used pseudo-devices.

File	Purpose
/dev/zero	File that produces a continuous stream of 0-byte characters
/dev/random	Random number generator
/dev/null	Special file that discards any data written to it

The Shell

The command line interpreter, also known as the shell, is a program that accepts input from a user (typically a command of some sort) and performs the requested task. Once that task has completed, the program will terminate and return the user to the shell.

The shell's prompt identifies the type of shell being used. There are two basic types of shell prompts:

- **$** Normal user shell (may also be **%** or **>** on some systems)
- **#** Root user shell

Below is the output of the **whoami** command (discussed on page 93) which highlights the difference between normal user and root shell prompts on the command line.

```
$ whoami
nick
```

Normal user shell

```
# whoami
root
```

Root user shell

Normal users typically have a limited amount of access to the system while the root user has unrestricted access. Section 5 covers this topic in more detail.

Warning	*The root user can do anything on Unix, Linux, and BSD systems. It is important to know when you are working as root to prevent accidental damage to your system.*

Many systems add customized information to the shell prompt. This information can be a handy indicator of things like the current user's login name, system host name, and the current directory as shown in the next example.

```
nick@mylaptop:/var/log $
```

Example of a customized shell prompt

The table below describes the information displayed in the customized shell prompt displayed in the previous example.

nick	@	mylaptop	:	/var/log	$
User Name	Spacer	Computer Name	Spacer	Current Directory	Shell Prompt

Types of Shells

There are several different types of shells for Unix, Linux, and BSD systems. Each shell has its own set of features and usage syntax. The next table describes the most popular shells that are currently available.

Shell	Prompt	Name	Notes
sh	$	Borne Shell	Default on some Unix systems
bash	$	Borne Again Shell	Enhanced replacement for the Borne shell Default on most Linux and Mac OS X systems
csh	%	C Shell	Default on many BSD systems
tcsh	>	T C Shell	Enhanced replacement for the C shell
ksh	$	Korn shell	Default on AIX systems

Types of shells

The **bash** shell is the most popular shell in use today. It is the default shell on all Linux systems and is also available for most Unix and BSD systems. Other shells like **csh**, **tcsh**, and **ksh** are typically used by default on non-Linux systems.

Accessing the Shell

When you first boot your system you are presented with a login prompt similar to the example pictured below.

```
Ubuntu Linux 9.04

Login: nick
Password: ******

$
```

After you enter your username and password the system will authenticate your login credentials via the `/etc/passwd` file. If the login information is correct, the system will start your shell (which is also specified in `/etc/passwd`). The **$** shell prompt indicates you are logged in and "ready to go."

Welcome to the world of the command line... Let's get started!